

Neutral Weak Current

Charged weak current:

- mediated by $W^\pm$
- does not conserve quark flavor
- conserves lepton flavor

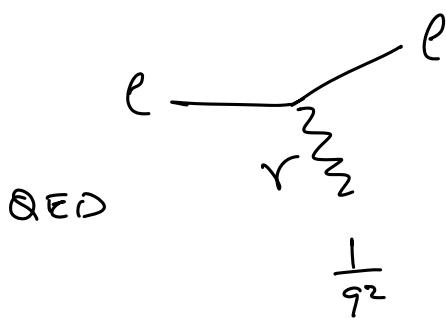
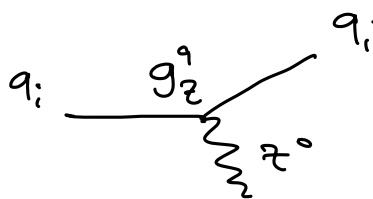
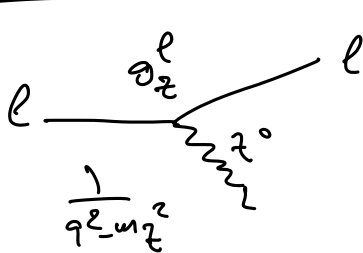
$$BF(K^0 \rightarrow \mu^+ \mu^-) \approx 10^{-9} \neq 0 \Rightarrow W^0 \text{ does not exist.}$$

$K^+ \rightarrow \pi^+ \nu \bar{\nu}$ not seen.

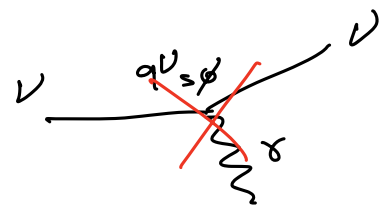


$\Rightarrow$ what about flavor conserving neutral weak current?

Signature of neutral current?



QED



not possible in QED

① $q^2 \ll m_Z^2$ QED dominates.

Indirect evidence for Z^0



No competition from QED

$\nu + \text{target} \rightarrow \nu + X$

Experimental challenges.

1) beam of neutrinos

2) β decay: ${}^A_Z X \rightarrow {}^A_{Z+1} X + e^- + \bar{\nu}_e$

Reines - Cowan (1956) near a reactor

$$\bar{\nu}_e + p \rightarrow n + e^+$$

$$\sigma \sim 10^{-43} \text{ cm}^2 \times \left(\frac{E_\nu}{\text{MeV}} \right)^2$$

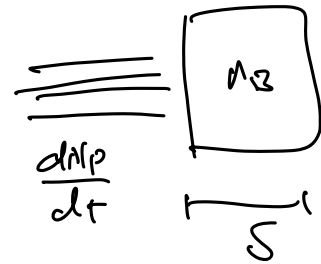
$$Q = m_{X_{A,Z}} - m_{X_{A,Z+1}} - m_e - m_\nu.$$

$$n \rightarrow p e^- \bar{\nu}_e \Rightarrow Q \approx 1.6 - 0.5 \approx 1 \text{ MeV}$$

binding energy for heavier nuclei: $\Rightarrow Q \approx \text{few MeV}$

ν from β decay have $E_\nu \lesssim \text{few MeV}$

$$\frac{dN_\nu}{dt} = \sigma \frac{dN_p}{dt} \cdot n_B \cdot \delta$$



$\Rightarrow$ intense and focused beam of neutrinos.

$\Rightarrow$ also increase E_ν as high as possible.

Experiment @ CERN 1973

$$\pi^\pm \rightarrow \gamma^\pm \nu_\mu / \bar{\nu}_\mu$$

$$\gamma^+ \nu_\mu / \gamma^- \bar{\nu}_\mu \approx 100\%$$

$$p + p \rightarrow p + p + \bar{u}^+ + \pi^-$$

$$p + \bar{p}$$

$$K^+ + K^-$$

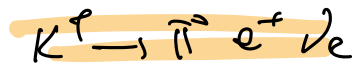
$$K^+ \rightarrow \mu^+ \nu_\mu$$

$$K^- \rightarrow \mu^- \bar{\nu}_\mu$$

$$\left. \begin{array}{l} K^+ \rightarrow \mu^+ \nu_\mu \\ K^- \rightarrow \mu^- \bar{\nu}_\mu \end{array} \right\} 65\%$$

Most of neutrinos
of μ flavor
 $\nu_\mu, \bar{\nu}_\mu$
dominating.

Some contamination of
 $\nu_e, \bar{\nu}_e$

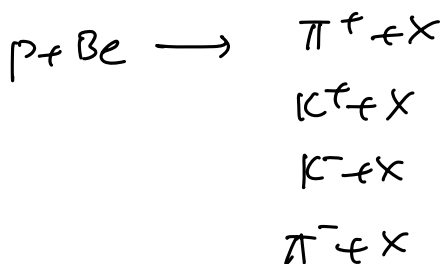


K^+ DECAY MODES	Fraction (Γ_i/Γ)	Scale factor/ Confidence level (MeV/c)	p
Leptonic and semileptonic modes			
$e^+ \nu_e$	$(1.582 \pm 0.007) \times 10^{-5}$		247
$\mu^+ \nu_\mu$	$(63.56 \pm 0.11) \%$	$S=1.2$	236
$\pi^0 e^+ \nu_e$	$(5.07 \pm 0.04) \%$	$S=2.1$	228
Called K_{e3}^+ .			
$\pi^0 \mu^+ \nu_\mu$	$(3.352 \pm 0.033) \%$	$S=1.9$	215
Called $K_{\mu3}^+$.			
$\pi^0 \pi^0 e^+ \nu_e$	$(2.55 \pm 0.04) \times 10^{-5}$	$S=1.1$	206
$\pi^+ \pi^- e^+ \nu_e$	$(4.247 \pm 0.024) \times 10^{-5}$		203
$\pi^+ \pi^- \mu^+ \nu_\mu$	$(1.4 \pm 0.9) \times 10^{-5}$		151
$\pi^0 \pi^0 \pi^0 e^+ \nu_e$	$< 3.5 \times 10^{-6}$	$CL=90\%$	135
Hadronic modes			
$\pi^+ \pi^0$	$(20.67 \pm 0.08) \%$	$S=1.2$	205
$\pi^+ \pi^0 \pi^0$	$(1.760 \pm 0.023) \%$	$S=1.1$	133
$\pi^+ \pi^+ \pi^-$	$(5.583 \pm 0.024) \%$		125

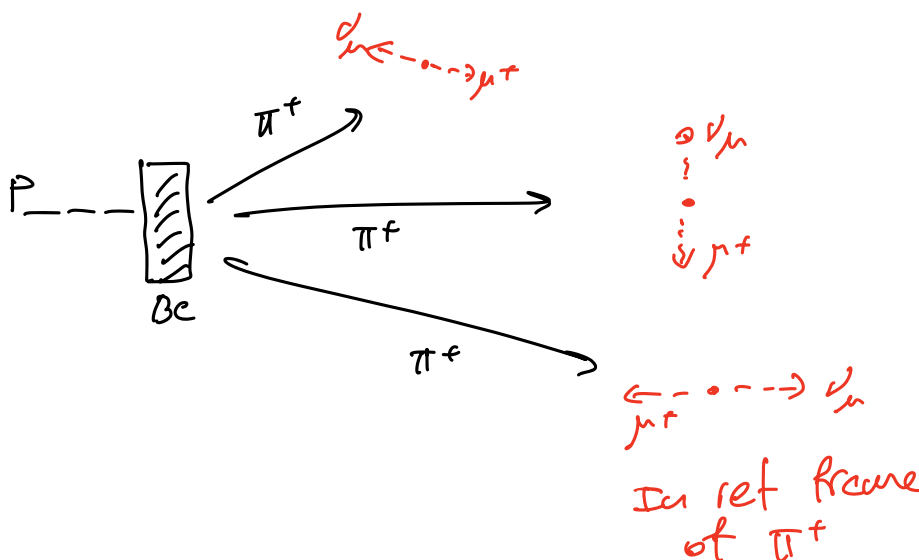
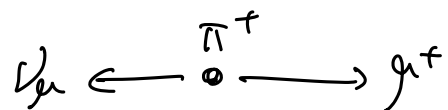
Magnetic field to separate π^+, K^+ from π^-, K^-



obtain pure beam of $\nu_\mu / \bar{\nu}_\mu$ and reject $\nu_e / \bar{\nu}_e$



In rest frame of π^+



beam of neutrinos
with diff. directions
and different
momentum.
unfocused beam

Exp problem: focus beam of neutrinos

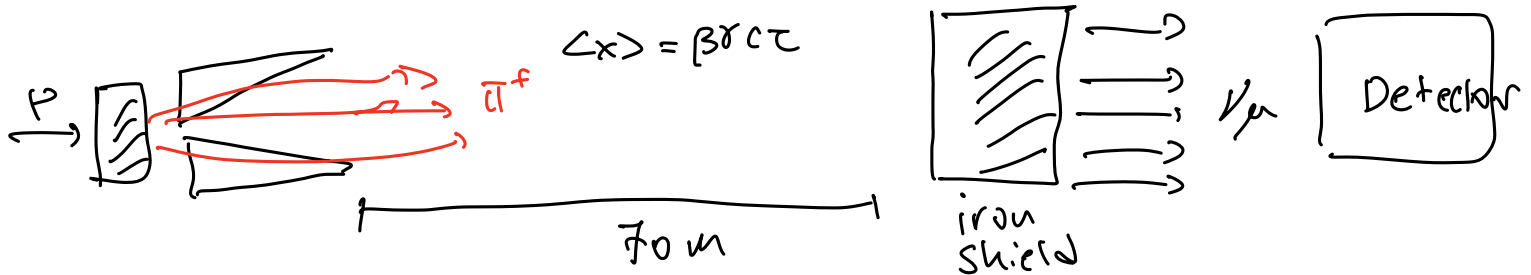
Solution: magnetic horn to focus pions
Simon van der Meer

$$\beta\gamma^* = \frac{E_\nu^* P_\mu}{m_\mu} = 0.25$$

$$E_\nu^{LAB} = \gamma_\pi (E_\nu^* + \beta_\pi P_\nu^*)$$

$$E_\nu^{LAB} > E_\nu^* = 25 \text{ MeV}$$

$$\sigma_{\nu\chi} \sim 10^{-43} \text{ cm}^2 \left(\frac{E_\nu}{\text{MeV}} \right)^2$$



$e^\pm$: EM shower, ionization

$\mu^\pm$: ionization (not stopped completely)